Propagation of Error: Citations to Retracted Research*

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Abstract

To shed light on how often scientists base their claims on problematic research, we exploit cases where problems with research are broadly publicized: retractions. We compare each of 8,627 retracted research articles, published 2000–2019 and retracted 2005–2022, with up to five articles from the same journal and publication year matched on pre-retraction citations, with citation histories through 2025 from OpenAlex for retracted and comparison articles alike. In the three complete years after the retraction year, citations to retracted articles were -62.1% (95% interval [-62.9, -61.2]%) relative to the path of their comparison articles, and the gap keeps widening for a decade. Retracted articles nonetheless continue to be cited: 26.8% of all citations to these articles arrived after the retraction year, five years on 39.3% of the articles observed that long were still being cited, and in a hand-coded sample of citing passages 234 of 256 post-retraction citations (91.4%) noted no concern with the cited article.

Keywords: citation behavior; retractions; scientific integrity; scientific misconduct; scientometrics

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1 Introduction

Citations are the bedrock of the scientific process. Scientists use citations to give credit for being first ("x, y, and z have studied a"), to debate methods and inferences ("the method used in study x fails to account for s"), as evidence ("x shows a", "we use data from x for our meta-analysis"), and to contextualize results ("our results are consistent with results from y"). And unless the researcher notes problems with cited research, citations cue that the data, results, inferences, or in some cases, the entire article, can be trusted.

When researchers cite articles with serious errors without acknowledging the errors, problems ensue. When erroneous research is cited as evidence (e.g., Chang et al., 2013; Torsvik et al., 2010), it cues that the evidence for the claim is good. Such citations unduly increase the reader's confidence in the result or argument. In the extreme, a reader may become persuaded that the point being buffeted by a citation to problematic research is right. The reader, generally another academic, may go on to write other articles influenced by the incorrect point, citing the erroneous article for support, or may share the point as fact with colleagues, students, and practitioners, propagating the error.

For example, consider an article published in April 2005 by Rubio et al. (2005) reporting a disturbing finding in *Cancer Research*. They report discovering that stem cells can spontaneously transform into cancerous cells during *in vitro* experiments. The finding was a blow to research in the use of stem cells to treat cancer. By the end of 2010, according to OpenAlex, the article had been cited 392 times.

In August 2010, the article was retracted (De la Fuente et al., 2010). The authors had been unable to replicate the result, and there was mounting evidence that transformations like the one reported were due to a basic error: cross-contamination during cell culturing. While this episode can be seen in a favorable light given that the errors were caught and a retraction notice was issued, the story does not end there.

Since 2010, the article has been cited another 429 times, with most citations noting no concern with the original work. For instance, a year after being retracted, Firinci et al. (2011) published an article in *International Immunopharmacology* citing Rubio et al. as basis for warning scientists that stem cells can spontaneously transform. Two years after the retraction, Kosaka et al. (2012) published an article in *Cancer Gene Therapy* in which they cited the evidence from Rubio et al. as a hurdle to implementation of the treatment they found to be effective. Three years after the retraction, Chang et al. (2013) published an article in *Aesthetic Plastic Surgery* citing Rubio et al. to argue that spontaneous transformation of human stem cells remains a risk.

Citing erroneous research without acknowledging errors is problematic in other ways. Such citations give full credit to research (and researchers) when at best partial credit is deserved. And since citation tallies cue credibility, such citations make erroneous research appear yet more credible.

Lastly, sometimes research with serious errors is cited to acknowledge the source of the data. For instance, Lin et al. (2013) used data from "two retracted studies ... without acknowledgment of their retractions, both of which were for fraudulent data..." (p. 1, Paul et al., 2015) in a meta-analysis. In such cases, the consequence is obvious and extreme—the key findings in the published work are incorrect.

In this paper, we study how much a retraction reduces citations to the retracted article, relative to what the article would otherwise have received, and how common citations to retracted research that do not note concerns with the original work are.

2 Citations to Retracted Research

The problem of citing problematic research without noting the problems takes special urgency in light of two empirical findings. First, the number of retractions—research where the problems are serious enough to warrant a retraction—is rising, even after controlling for the number of articles published (Grieneisen and Zhang, 2012; Steen, 2011; Oransky, 2022; Van Noorden, 2023).[1] Second, the main reasons research is retracted—major error and fraud—are concerning (Bozzo et al., 2017; Grieneisen and Zhang, 2012; Singh et al., 2014; Fang et al., 2012; Wager and Williams, 2011).

Because of these reasons, researchers have started to look at how citation rates change post retraction (Bar-Ilan and Halevi, 2018; Rubbo et al., 2019) and whether citations post retraction acknowledge problems in the retracted work (e.g., Bar-Ilan and Halevi, 2017; Hamilton, 2019; Luwel et al., 2018). Bar-Ilan and Halevi (2018), for instance, use a sample of 994 retracted articles from `ScienceDirect` from a single month—October 2014—and find that the total number of citations to retracted articles grew over time. Similarly, Rubbo et al. (2019) use a sample of 238 retracted engineering articles and their citations and report the number of times the articles are cited pre and post retraction and find no apparent difference. Neither article notes whether citations after retraction note problems with the cited article. Some studies, however, consider whether post retraction citations acknowledge problems. Bar-Ilan and Halevi (2017), for instance, find that, of the 283 citations to retracted case-study articles, a majority did not note problems with the article they cited. Similarly, Hamilton (2019) report that of 358 citations to retracted articles in the radiation oncology field, 92% referenced the research as legitimate. Finally, Luwel et al. (2018) studied pre- and post- retraction citations to a set of highly publicized retractions and found that over 95% of citations were positive or neutral in how the researchers characterized the cited work. Older research has found similar results. A study using a database of 235 retracted biomedical articles found that nearly 94% of the citations after retraction treated research as valid (Budd et al., 1998) and work done as far back as 1990, exploiting a dataset of 82 retracted articles, came to similar conclusions (Pfeifer and Snodgrass, 1990). The same picture recurs in single cases followed for a decade or more (Bornemann-Cimenti et al., 2016; Schneider et al., 2020; Suelzer et al., 2019; Heibi and Peroni, 2021), in fields from dentistry to radiology to psychology (Theis-Mahon and Bakker, 2020; Rapani et al., 2020; Bolboacă et al., 2019; Morís Fernández et al., 2019; Dinh et al., 2019), in systematic reviews and clinical guidelines that rest on retracted trials (Avenell et al., 2019; Bolland et al., 2021; Kataoka et al., 2022), in policy documents (Malkov et al., 2023), and in a recent census of citations to retracted biomedical papers, in which two in five post-retraction citations use the retracted article as support and almost none flag the retraction (Joseph, 2026).

[1] Steen et al. (2013) posits that the rise in retractions is a result of declining quality of research and better ability to detect problems. Part of the recent rise is the discovery of paper mills (Else and Van Noorden, 2021; Candal-Pedreira et al., 2022).

Citing authors who are unfamiliar with a field, or who rely on prominent journals, are the most likely to cite retracted work (Hsiao and Schneider, 2021; Woo and Walsh, 2024).

While these studies consistently show high rates of positive or neutral citations to retracted research following retraction, most use small samples or samples that span a single discipline. The use of small, selective samples means that we cannot confidently say how widespread the problem is. The work also mostly describes citations after retraction rather than the change in citations that the retraction causes, which needs a comparison: citations to any article rise for a few years after publication and then fall, and most retractions happen two to three years after publication, near the peak, so a before-and-after comparison confounds the retraction with the ordinary life cycle of citations. The studies that build such a comparison find large penalties. Pfeifer and Snodgrass (1990) already reported a control-group decline of about a third. With articles from the same journal issue as controls, Furman et al. (2012) find that citations to retracted biomedical articles fall by more than half; with controls from the same field and year matched on the pre-retraction citation path and a Poisson model with article fixed effects, Lu et al. (2013) find declines of more than four fifths five or more years after retraction, and a penalty that spreads to the retracted authors' other work; Shuai et al. (2017) match on journal, publication date and pre-retraction citations and find post-retraction citations halved, most for proven misconduct; and Kühberger et al. (2022) compare retracted PubMed articles with randomly drawn articles from the same journal and year and find a drop of roughly three fifths. Memon et al. (2026) match Retraction Watch retractions to articles from the same venue, year and discipline with the same citations at retraction and show that the most visible articles lose the most. Related work follows the penalty to the retracted authors' other articles and their coauthors (Azoulay et al., 2015, 2017; Jin et al., 2019; Mongeon and Larivière, 2016; Hussinger and Pellens, 2019), to the articles that cite retracted work (Wang et al., 2026; García-Romero et al., 2026), to online attention (Peng et al., 2022), and to clinical trial reports (Mott et al., 2019; Song et al., 2025); Xu et al. (2023) show that coverage by Retraction Watch itself reduces post-retraction citations.

In this paper, we bridge these gaps with a census of eligible retractions across fields rather than a biomedical sample, open data so the result can be rebuilt, a comparison that reports its own pre-trends, heterogeneity and sensitivity, and the descriptive level of continued citation set beside the causal penalty. The companion paper examines the same questions for a serious statistical error that does not lead to retraction (Cor and Sood, 2026).

3 Hypotheses and Research Design

We expect the publication of the retraction notice to raise awareness about the retracted article, which should reduce citations to it, especially citations that fail to acknowledge the problem. Retractions are unequivocal indicators of serious problems with an article, they elicit a response from the publishers, who often switch titles of the retracted articles in their online databases to reflect that they have been retracted, and they are tied to specific articles. Formally, we hypothesize that retracted articles will receive fewer citations per year after the retraction than they would have received had they not been retracted.

Estimating that requires a comparison. We follow the matched-control design of Furman et al. (2012) and Azoulay et al. (2015): each retracted article is paired with articles from the same journal and publication year with similar citations before the retraction, and the outcome is the retracted article's proportional change relative to its own comparison articles. Each retracted article and its comparison articles form a stack. The main estimate is a Poisson quasi-maximum-likelihood regression of annual citations on an indicator for a retracted article in the post-retraction years, with a fixed effect for each article within its stack and for each stack and event year, so every stack supplies its own citation path and the coefficient is the retracted article's proportional change relative to its matched articles (Wooldridge, 2023; Chen and Roth, 2024; Bergé, 2018). The pre-period is up to three years before retraction and the post-period is years one to three after; the retraction year is reported separately because it is only partly exposed. Comparison articles receive weight one over the number of matches in their stack. Standard errors are clustered by article, because a comparison article can serve several stacks.

The dynamic version replaces the indicator with event-year interactions, with the year before retraction as the reference, which displays the pre-trend and the shape of the response. The figures show every event year the data hold, from ten years before retraction to ten after. Each stack contributes the years that lie between its publication year and 2025, and because the comparison articles share the retracted article's publication year, both arms of a stack cover the same calendar years; what changes across event years is which stacks contribute, so the number of retracted articles behind each point is printed beneath it. The early pre-retraction years come only from articles retracted long after publication and the late post-retraction years only from early retractions. A second comparison drops the citation matching and uses every sampled article from the same journal and publication year with equal weights; it asks whether the matching step, rather than the journal-year comparison, drives the estimate. An additive version estimates the same contrast in citations per year. A synthetic difference-in-differences estimate by retraction-year cohort (Arkhangelsky et al., 2021) re-weights the comparison articles and the pre-years to fit the retracted articles' pre-period path, for the stacks with three complete years on each side, with whole-article bootstrap intervals. The heterogeneity splits are reason, field, pre-retraction citation tercile, years to retraction, and a prior expression of concern.

4 Data

4.1 Retracted articles

The retraction registry is the Retraction Watch database, which Crossref publishes as an open file (Retraction Watch and Crossref, 2026; Hendricks et al., 2023); among the multidisciplinary sources it is the most complete index of retractions, and the sources disagree with one another more than one would hope (Schneider et al., 2023; Ortega and Delgado-Quirós, 2024). Our snapshot holds 67,440 retraction records. We keep one record per original DOI (the earliest retraction), require the original to be a research article or clinical study published 2000–2019 and retracted 2005–2022 at least two years after publication, so that at least one complete pre-retraction year exists, and require the DOI to resolve to one OpenAlex journal article whose publication year agrees with

Retraction Watch's within a year. Table A1 in the appendix lists every step. The frame has 9,555 articles, of which 8,963 resolve.

4.2 Citations

The outcome is the number of articles and reviews citing an article in each year, from the article's publication year through 2025, from OpenAlex (Priem et al., 2022), whose reference coverage now approaches that of Web of Science and Scopus (Culbert et al., 2025; Thelwall and Jiang, 2025). We take retraction status from Retraction Watch rather than OpenAlex's own flag, which has been unreliable (Hauschke and Nazarovets, 2025). Years with no citations are zeros, not missing. Comparison articles are measured with the same query. We validated the measure against a 2016 Web of Science export of citations to the same articles, described in Appendix E: for 19,351 article-years covering the same articles from publication through 2015, OpenAlex counts are 1.10 times the Web of Science counts, agree exactly in 50.0% of article-years and within one citation in 78.2%. For the 1,284 articles in both the 2016 sample and the frame, the Retraction Watch retraction year equals the Web of Science notice year in 94.2%.

4.3 Comparison articles

For each retracted article we draw a seeded random sample of twenty articles from the same OpenAlex journal and publication year, excluding any that OpenAlex or Retraction Watch flags, and match on two quantities measured strictly before the retraction year: the logarithm of one plus cumulative citations, and the logarithm of one plus citations in the year before retraction. Both are standardized within the pool, a candidate must be within one standard deviation on each, and the five nearest are kept with equal weight. 8,627 retracted articles find at least one comparison article (26,138 distinct comparison articles); 323 have no candidate within the caliper and are reported, not dropped. Matching uses no information from the retraction year onward, and a test in the repository perturbs post-retraction outcomes and checks that the matches do not change.

4.4 Acknowledgment coding

To estimate how many of the citations after retraction fail to acknowledge problems, in 2016 we coded a random sample of 100 articles that cited a retracted article before or in the year of the retraction notice and 275 articles that cited retracted research a year or more after the publication of the retraction notice. Appendix F describes the coding.

5 Results

5.1 Retractions

Retraction Watch's reasons are grouped literally: 41.0% of the frame articles carry a fabrication, manipulation, paper-mill or peer-review label, 32.8% a plagiarism or duplication label, and 18.9% an error or unreliability label; the rest state an investigation or little else. The share attributable to misconduct is similar to the number obtained by other research on reasons for retraction in other

Table 1: Citations to retracted articles by years after the retraction year, for the retracted articles in the frame with a complete OpenAlex history. Each row covers the articles whose retraction was at least that many years before the end of the citation data (2025). Cited at all is the share of those articles with at least one citation in that year.

Years after retraction	Articles observed	Citations	Mean	Median	Cited at all (%)	Five or more (%)
1	8,963	21,144	2.4	1	61.1	13.9
2	8,963	15,214	1.7	1	52.1	9.3
3	8,963	11,388	1.3	0	44.8	6.1
4	7,425	8,493	1.1	0	42.0	5.2
5	5,821	6,239	1.1	0	39.3	4.6
6	4,882	4,679	1.0	0	36.1	4.2
7	4,150	3,696	0.9	0	34.9	3.4
8	3,531	2,763	0.8	0	31.6	3.2

corpora. For instance, a study of 1,112 biomedicine articles retracted between 1997 and 2009 found that 55% were retracted for some type of misconduct (Budd et al., 2011) (see also Steen (2011); Fang et al. (2012)). Articles are mostly retracted because the research cannot be trusted.

These flawed articles often accrue a fair number of citations before anything is public. It took, on average, 5.0 years for an article to be retracted. The median time before the article was retracted was 4 years (see Figure A3 in the appendix) with 55.5% of the articles taking four or more years to be retracted. These numbers are similar to those obtained elsewhere (e.g., Steen et al., 2013). The number of retractions per published article has also risen over the period, from under one per ten thousand articles published in 2000 to over two per ten thousand published in 2018 (Figure A2 in the appendix); the pattern is consistent with Steen et al. (2013), who also find a rapid increase in retractions over time.

5.2 Retracted articles keep being cited

Given that most retracted articles are retracted because of serious error or fraud, we expect retracted articles to *never* be cited without acknowledgment of the problem a year or more—taking account of long publication windows—after the retraction notice has been published. However, for the 8,963 retracted articles in the frame with a complete OpenAlex history, 82,013 of 305,552 citations, 26.8%, arrived in the years after the retraction year. 80.4% of the articles observed for at least one full year after retraction were cited at least once in those years; the median such article collected 3 post-retraction citations and the mean 9.2. Given the skew in retractions, with the bulk being published in recent years, these totals include very little post retraction data for many of the articles, so they are a *lower bound* of the percentage of citations that happen after an article has been retracted. Table 1 follows the articles year by year: five years after retraction, 39.3% of the 5,821 articles observed that long were still cited that year, at a mean of 1.1 citations, and eight years after retraction 31.6% were. These are counts of citing articles and reviews in OpenAlex, not a judgement of how each citation was used; Section 5.7 reports the sample in which the citing passages were read.

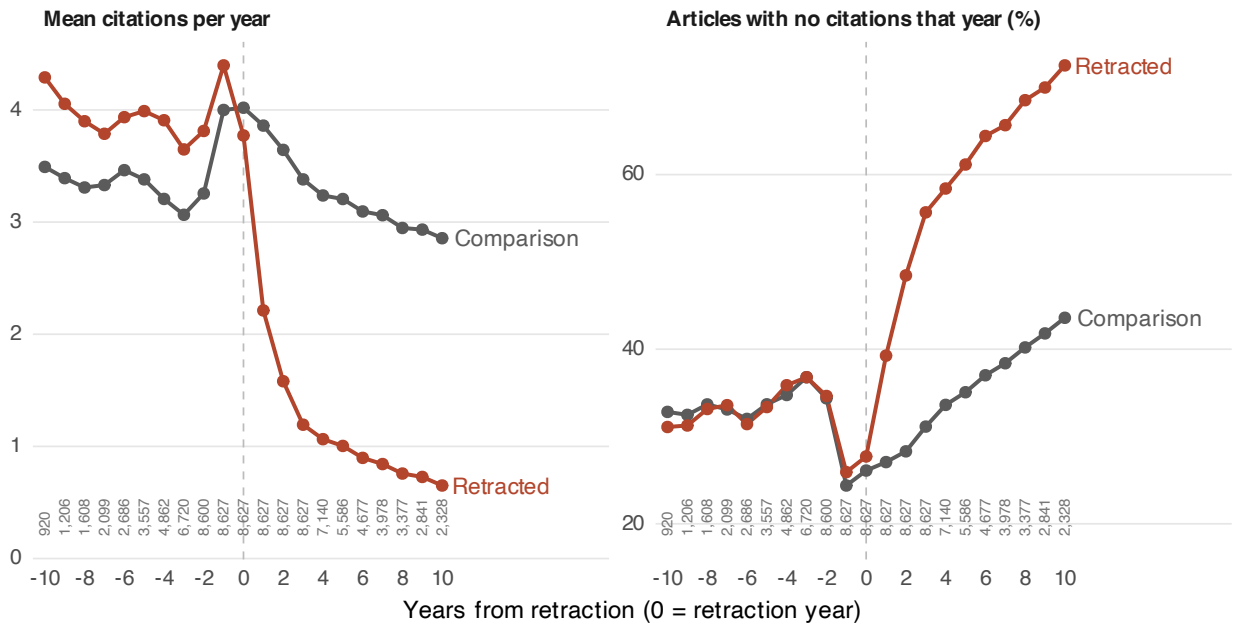


Figure 1: Citations per year to retracted articles and their matched comparison articles, by years from retraction. Means over articles and the share of articles with no citations that year, weighting each comparison article by one over the number of matches in its stack; the comparison articles come from the same journal and publication year as the retracted article and were matched on citations before the retraction year. The number of retracted articles contributing to each year is printed along the bottom. No intervals are drawn: the panel is a census of the matched articles.

5.3 Citations fall after retraction and keep falling

To explore the frequency of citations before and after retraction, Figure 1 plots mean citations per article per year, and the share of articles with no citations that year, against years from retraction, for retracted and comparison articles. Before retraction the two paths have the same shape, rising toward the year before retraction, and the retracted articles sit a little above their matches throughout: the nearest neighbours of a well-cited article among twenty sampled candidates lie mostly below it, and the fixed effects absorb that level difference. In the year before retraction the mean retracted article received 4.4 citations and the mean comparison article 4.0. Three years after retraction the means were 1.2 and 3.4, and 55.7% of retracted articles received no citations that year, against 31.1% of comparison articles.

The main model puts the change at -62.1% (95% interval [-62.9, -61.2]%) over years one to three after retraction (Table 2). Extending the post-period to five years gives -65.8% [-66.6, -65.0]%. In citations per year, the additive model gives a change of -2.45 [-2.55, -2.36] citations per year. Replacing the matched articles with all 81,796 sampled articles from the same journal and publication year, unmatched and equally weighted, gives -63.1% [-63.7, -62.4]%: the journal-year comparison carries the result, and the matching step changes it little. The event-year estimates in Figure 2 show the time path: 4.2% and 4.1% in the two years before the reference year; -18.5% in the retraction year; -51.5% one year after; -74.8% [-75.8, -73.8]% five years after; and the gap is still widening ten years after retraction among the articles observed that long.



Figure 2: Change in citations to retracted articles relative to comparison articles, by years from retraction: matched comparison articles (dark) and all sampled articles from the same journal and publication year, unmatched (grey). Poisson coefficients with article-within-stack and stack-by-year fixed effects, converted to percent; the year before retraction is the reference (open circles). Bars are 95% intervals from standard errors clustered by article. The number of retracted articles contributing to each year is printed along the bottom.

The pre-period coefficients are small but not zero. The ratio of retracted to comparison citations was higher three and two years before retraction than in the year before, so the retracted articles were already slipping relative to their matches, and a joint test of the two pre-period coefficients rejects equality ($p < 0.001$). Without matching the pre-period gap is larger (6.3% and 5.9%), which is what the matching on pre-retraction citations is for. The sample is large enough to detect a gap of a few percent, and this one is in the direction of the later decline: whatever makes an article retractable may already cost it citations before the notice. We therefore report the sensitivity bounds of Rambachan and Roth (2023), which allow the post-retraction gap to depart from the comparison path by a multiple of the largest year-to-year movement seen before retraction (4.1%). Allowing departures up to 2.0 times that movement, the three-year change stays within [-70.0, -53.1]%. A pre-trend of the size we observe cannot produce a decline of the size we estimate.

5.4 Synthetic difference-in-differences

Within each retraction-year cohort, the synthetic difference-in-differences estimate re-weights the comparison articles to match the retracted articles' pre-period path. Pooled by inverse variance over 18 cohorts and 6,720 retracted articles with three complete years on each side, the change is -2.27 citations per year (95% interval [-2.41, -2.14]); the cohort estimates are in Table A3 in the appendix.

Table 2: Change in citations to retracted articles relative to matched comparison articles. PPML rows report the percent change implied by the Poisson coefficient; the OLS row reports citations per year. Intervals are 95%, clustered by article.

Specification	Model	Retracted / comparison	Estimate	95% interval
Main: matched stacks, years +1 to +3	PPML	7,567 / 23,528	-62.1%	[-62.9, -61.2]%
Years +1 to +5	PPML	7,650 / 23,912	-65.8%	[-66.6, -65.0]%
Year +1 only	PPML	7,198 / 21,792	-51.7%	[-52.8, -50.6]%
Pre-period -1 only	PPML	7,228 / 22,870	-62.3%	[-63.1, -61.4]%
Additive (OLS)	OLS	8,627 / 26,138	-2.45	[-2.55, -2.36]
Nearest match only	PPML	7,312 / 5,943	-60.4%	[-61.8, -59.0]%
All sampled journal-year articles, unmatched	PPML	7,929 / 81,796	-63.1%	[-63.7, -62.4]%

5.5 Who is penalized

Table 3 reports the main contrast by reason for retraction, field, pre-retraction citation tercile, years to retraction, and a prior expression of concern, with a joint test of equality within each split. The decline is large in every group and differs by group. Articles retracted for misconduct (-64.2%) or error (-64.1%) lose more than articles retracted for plagiarism or duplication (-57.8%), and articles whose notice states no clear reason lose least (-47.4%; joint test $p < 0.001$). Across fields the change runs from -67.7% to -56.0% ($p < 0.001$). Articles retracted sooner lose more: -67.1% when retracted two years after publication against -56.0% after six or more years ($p < 0.001$), consistent with a penalty that is proportional to citations the article would still have been collecting. Articles that carried an expression of concern before the retraction (Vaught et al., 2017; Grey et al., 2022) fall by -66.7%, against -61.7% for the rest ($p = 0.003$). The pre-retraction citation terciles differ little ($p = 0.075$).

5.6 Robustness

Table 4 drops each retraction-year cohort, each field, journal-years with more than ten retractions, articles with a prior expression of concern, and the top one percent of articles by pre-retraction citations, and clusters by journal instead of article. The leave-one-out estimates run from -63.0% to -60.4%. Clustering by journal rather than article widens the interval to [-63.7, -60.4]%.

5.7 Post-retraction citations rarely acknowledge the retraction

We could not locate 32 of the sampled citing articles, leaving us with 343 articles. There were no false positives. Of the 87 articles citing retracted articles before or in the same year the retraction notice was published, 85 (97.7%) did not acknowledge any problems with the research being cited. Of the 256 articles citing retracted articles a year or more after the retraction notice was published, 234 (91.4%, 95% Wilson interval [87.3, 94.3]%) did not mention problems—a slight trend downward. Table 5 gives the full accounting. In all, the data suggest that retracted articles continue to be cited at non-zero rates after retraction and that the citations are still very much failing to acknowledge concerns.

Table 3: Change in citations relative to matched comparison articles, years one to three after retraction, by group. Each block is one Poisson model with group-specific post-retraction coefficients; the p-value is a joint Wald test of equal coefficients within the block. Intervals are 95%, clustered by article.

Group	Retracted articles	Change (%)	95% interval
Field (joint test p <0.001)			
Life sciences	4,628	-62.3	[-63.3, -61.4]%
Technology and environment	645	-60.0	[-63.4, -56.3]%
Health sciences	1,253	-56.0	[-58.8, -53.0]%
Physical sciences	1,400	-66.4	[-69.0, -63.7]%
Social sciences and humanities	184	-67.7	[-71.9, -62.9]%
Years to retraction (joint test p <0.001)			
2	1,836	-67.1	[-69.2, -64.7]%
3	1,740	-65.9	[-67.6, -64.1]%
4–5	2,045	-63.1	[-64.5, -61.7]%
6 or more	2,489	-56.0	[-57.4, -54.6]%
Pre-retraction citations (tercile) (joint test p 0.075)			
Lowest	1,998	-59.3	[-62.2, -56.1]%
Middle	3,175	-61.2	[-62.4, -60.0]%
Highest	2,937	-62.5	[-63.5, -61.4]%
Prior expression of concern (joint test p 0.003)			
No	7,771	-61.7	[-62.6, -60.8]%
Yes	339	-66.7	[-69.6, -63.6]%
Reason for retraction (joint test p <0.001)			
Plagiarism or duplication	2,586	-57.8	[-59.3, -56.2]%
Error	1,637	-64.1	[-65.7, -62.3]%
Misconduct	3,405	-64.2	[-65.3, -63.1]%
Other or unstated	482	-47.4	[-51.6, -42.9]%

6 Discussion

Only some of the points in a scientific paper rely on original work. For most points, scientists rely on work done by others. For instance, scientists rely on other research to buffet arguments, situate work in a research tradition, credit others for original arguments, data, and results, etc. (Erikson and Erlandson, 2014). However, sometimes, the work done by others is problematic. For instance, sometimes, the claims in the cited work are not only wrong but made with fraudulent data. And we expect scientists to either not refer to such work or acknowledge the issues fully, especially after the errors have been publicized.

In this paper, we use a census of retracted research articles, their citations, and matched comparison articles, and discover that retraction works, partly. Relative to comparable articles, a retracted article loses a large share of the citations it would have received, and the loss grows over the following years. But the retracted article is not forgotten: 26.8% of the citations these articles have ever received came after the retraction year, five years on 39.3% of them were still being cited, and in the hand-coded sample almost none of the citing passages mentioned the retraction. Both

Table 4: Main contrast under leave-one-out and alternative clustering. Each row refits the main model on the stated subsample. Intervals are 95%.

Specification	Retracted articles	Change (%)	95% interval
Drop retraction year 2005	8,088	-62.1	[-62.9, -61.2]%
Drop retraction year 2006	8,054	-62.1	[-62.9, -61.2]%
Drop retraction year 2007	8,058	-62.2	[-63.0, -61.4]%
Drop retraction year 2008	8,016	-62.0	[-62.8, -61.1]%
Drop retraction year 2009	7,973	-62.2	[-63.1, -61.4]%
Drop retraction year 2010	7,971	-62.0	[-62.9, -61.2]%
Drop retraction year 2011	7,845	-62.1	[-63.0, -61.2]%
Drop retraction year 2012	7,818	-62.2	[-63.1, -61.4]%
Drop retraction year 2013	7,790	-62.3	[-63.2, -61.5]%
Drop retraction year 2014	7,791	-62.1	[-63.0, -61.3]%
Drop retraction year 2015	7,677	-61.9	[-62.8, -61.1]%
Drop retraction year 2016	7,619	-62.7	[-63.5, -61.8]%
Drop retraction year 2017	7,612	-62.2	[-63.0, -61.3]%
Drop retraction year 2018	7,546	-62.2	[-63.1, -61.3]%
Drop retraction year 2019	7,473	-62.4	[-63.3, -61.5]%
Drop retraction year 2020	7,250	-62.1	[-63.0, -61.3]%
Drop retraction year 2021	6,630	-61.7	[-62.6, -60.8]%
Drop retraction year 2022	6,659	-60.4	[-61.3, -59.4]%
Drop field: life sciences	3,482	-61.6	[-63.2, -59.9]%
Drop field: technology and environment	7,465	-62.2	[-63.0, -61.3]%
Drop field: health sciences	6,857	-63.0	[-63.9, -62.1]%
Drop field: physical sciences	6,710	-61.3	[-62.2, -60.5]%
Drop field: social sciences and humanities	7,926	-62.0	[-62.8, -61.1]%
Drop journal-years with more than ten retractions	6,118	-61.9	[-62.8, -60.9]%
Drop articles with a prior expression of concern	7,771	-61.7	[-62.6, -60.8]%
Drop top 1% by pre-retraction citations	8,023	-61.7	[-62.5, -60.8]%
Journal-clustered standard errors	8,110	-62.1	[-63.7, -60.4]%

Table 5: Hand coding of whether a citing passage acknowledged a problem with the cited retracted article, by whether the citing article appeared before, in, or after the year of the retraction notice. Intervals are 95% Wilson intervals for the share with no problem acknowledged.

Citing year	Sampled	Not located	Coded	No concern recorded	Share (%)	95% interval
Before the retraction year	58	6	52	52	100.0	[93.1, 100.0]
Retraction year	42	7	35	33	94.3	[81.4, 98.4]
After the retraction year	275	19	256	234	91.4	[87.3, 94.3]

facts can hold at once. A penalty is a comparison with a counterfactual; continued citation is a level.

The causal reading requires that the matched comparison articles follow the citation path the retracted article would have followed absent retraction. Matching on journal, year and pre-retraction citations makes that assumption plausible but the pre-period coefficients show it is not exact: retracted articles were already slipping a few percent a year relative to their matches before the notice. Articles are not retracted at random, and an article under investigation may lose citations

before anything is public, so part of the measured decline may belong to the events that led to the retraction rather than to the notice itself. The sensitivity bounds say how much that matters: a continuation of the pre-retraction slippage, even at twice its observed pace, leaves most of the decline in place. Reason labels are a notice's stated grounds, not an adjudication, and the acknowledgment sample is small and dated.

We have a few hypotheses (but no data) about why these types of citations exist. One shallow reason is that journal publishers do not seem to screen citations to retracted research in submitted manuscripts. Instead, they assume that scientists will cite appropriately and that the peer review process will screen the remaining problems.

Another potential reason is that scientists do not scrutinize the articles they cite. One reason for that is that they trust published work. Their trust is likely buffeted by the belief that scientific misconduct is limited to a few bad people, which in turn may be driven by the fact that media reporting often focuses on personalities rather than processes. For example, cases of Diederik Stapel, who fabricated data behind at least 30 papers (Levelt et al., 2012), John Darsee, who faked data behind nearly 100 publications (Stewart and Feder, 1987; Anderson et al., 2013; Wallis, 1983), and Jan Hendrik Schön, who during a period in 2001 published a research paper every eight days based on fabricated data (Service, 2003; Anderson et al., 2013) were heavily covered in the media. So were the cases of Andrew Wakefield, who published an article linking MMR vaccine to autism using fabricated data (Wakefield et al., 1998; Deer, 2011; Godlee et al., 2011), and Michael LaCour, who published a paper in *Science* based on fabricated data (Broockman et al., 2015; McNutt, 2015).[2] Each of these cases was framed as an example of misconduct by a bad actor, the subtext often being that a bad actor is an outlier.

Misconduct, however, is not limited to a few bad actors. A large anonymous survey of early- and mid-career scientists found that about 2% of scientists admitted to engaging in fabricating, falsifying, or plagiarizing in the last *three* years (Martinson et al., 2005) (see also Titus et al. (2008)). Another study found that nearly 34% of the respondents in past surveys admitted to engaging in questionable research practices (Fanelli, 2009).

The other likely reason for trust in peer-reviewed research is that the rate of official retractions is extremely low. For instance, in a study of the nearly 9.4 million articles published between 1950 and 2004 and available on PubMed, only 596 had been retracted (Cokol et al., 2007). In all likelihood, however, the actual rate of serious errors in manuscripts is manifolds that rate. For instance, Cokol et al. (2007) estimate the rate at which articles ought to be retracted to be anywhere between 16.7 times to 167.8 times the actual rate. And these estimates do not account for research that involves harder-to-prove malpractice such as filing away non-significant results (Franco et al., 2014), conducting specification searches, and other more fundamental concerns like low power, which reduces the likelihood that a nominally statistically significant finding reflects a true effect (Button et al., 2013; Ioannidis, 2005).

It is also likely that scientists cite erroneous research due to a lack of time stemming from the pressure to publish. Some scientists may also buckle to incentives to treat published research

[2]Other prominent cases include that of William Summerlin, who painted mice rather than transplant skin (Basu, 2006; Anderson et al., 2013), Woo Suk Hwang, who claimed to have cloned embryos, Eric Poehlman, who fabricated data behind at least ten papers and numerous grant applications.

generously, given harsh (but accurate) judgments may provoke some reviewers. More generally, peer reviews tend to focus on instances where the authors fail to cite someone or miss the journal's formatting requirements more than concerns over incorrectly citing bad research. This means that there is little incentive to cite carefully.

Static reference databases likely exacerbate the problem. While publisher databases are quick to mark papers as retracted, many researchers rely on static reference databases stored on their computers for citations. These databases sometimes contain articles that have since been retracted. For instance, Davis (2012) finds that personal Mendeley libraries contained 1,340 retracted articles. In all, there are many reasons why scientists may cite erroneous research, even after the errors have been publicized via a retraction notice.

These 'miscitations' are avoidable. Improving how the information about problematic research is generated and communicated can ameliorate the problem. The retraction registry we used is open (Retraction Watch and Crossref, 2026), and checking a reference list against it is a few lines of code. Publishers can screen submitted manuscripts for citations to retracted research. Web browser add-ons for scholarly search engines such as Google Scholar and JSTOR can highlight problematic articles listed on a web page. Tools that automatically create pull requests to personal bibliography libraries posted on open publication platforms like GitHub could be built. Taking steps like these should lead to significant improvements in reducing citations of problematic research.

To conclude, this study contributes to a rich literature on scholarly citation behavior. For example, Rekdal (2014) introduces the concept of *academic urban legends*—rumors that appear frequently and in complex and colorful ways that can usually be traced back to poorly employed sources—like the story of a decimal point error resulting in the belief that spinach is a good source of iron. Similarly, Lance et al. (2006) explores the basis for *methodological urban legends* in the form of four commonly employed criteria for the goodness of fit indices, reliability, interrater agreement indices, and eigenvalues. They find that much of the hype about these criteria has emerged from an inappropriate citation of the source material. By the same token, Harzing (2002) examined the citation network of 60 references on expatriate failure rates (EFRs)—the phenomenon of an expatriate returning home before their contractual period of employment expires. He finds that one reference was cited 22 times, with only six correctly representing the EFRs reported in the original work. Similarly, Todd et al. (2010) report that of 198 randomly selected citations to articles in two recent issues of 33 marine biology journals, 1 in 4 represented an inappropriate citation—ambiguous, no support, or empty (citation to a secondary source).

We find that citations to retracted research fall sharply relative to comparable research once the retraction is published, and keep falling. We also find that retracted research continues to be cited for years afterward, and that the rate at which researchers cite it without acknowledging problems remains high.

Statements and declarations

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Data availability. Every dataset, script and generated number is in the public repository at `https://github.com/recite/propagation_of_error`. The retraction registry is the Retraction Watch database as distributed by Crossref, and citation histories were retrieved from the OpenAlex API; the repository holds the retrieved data and a receipt for every retrieval. The 2016 Web of Science exports and the hand coding are included as retrieved.

Author contributions. Both authors designed the study, collected the data, analysed it and wrote the paper.

Use of AI. Code, data processing and drafts of the text were prepared with the assistance of a large language model (Claude, Anthropic); the authors specified the design, checked every analysis and number, and are responsible for the content.

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Table A1: From the Retraction Watch snapshot to the analysed stacks.

Step	Articles
Retraction Watch retractions	67,440
Frame: research articles, published 2000-2019, retracted 2005-2022, two or more years after publication, with a DOI	9,555
Resolved to an OpenAlex journal article in the same year	8,963
Matched to at least one comparison article	8,627
With at least one complete post-retraction year	8,627
With three complete pre- and post-retraction years (synthetic DiD)	6,720

Table A2: Event-year coefficients from Figure 2, in percent, for the matched comparison and for the unmatched journal-year comparison.

	Matched comparison		Journal-year, unmatched	
Years from retraction	Change (%)	95% interval	Change (%)	95% interval
-10	12.5	[6.3, 19.0]%	25.7	[19.5, 32.2]%
-9	8.1	[3.0, 13.4]%	18.6	[13.5, 23.8]%
-8	5.9	[1.0, 11.0]%	15.0	[10.5, 19.5]%
-7	1.2	[-2.9, 5.5]%	8.6	[4.9, 12.3]%
-6	0.0	[-3.7, 3.9]%	7.8	[4.7, 11.0]%
-5	1.7	[-1.8, 5.3]%	9.6	[6.8, 12.4]%
-4	4.9	[1.9, 8.1]%	8.9	[6.4, 11.3]%
-3	4.2	[1.7, 6.8]%	6.3	[4.3, 8.2]%
-2	4.1	[2.2, 6.0]%	5.9	[4.3, 7.6]%
-1	reference	–	reference	–
0	-18.5	[-20.1, -16.9]%	-18.4	[-19.6, -17.2]%
1	-51.5	[-52.6, -50.3]%	-51.3	[-52.2, -50.4]%
2	-63.5	[-64.5, -62.5]%	-64.0	[-64.7, -63.2]%
3	-70.5	[-71.4, -69.6]%	-71.4	[-72.1, -70.7]%
4	-73.2	[-74.1, -72.2]%	-73.8	[-74.5, -73.1]%
5	-74.8	[-75.8, -73.8]%	-75.5	[-76.3, -74.8]%
6	-77.4	[-78.4, -76.3]%	-78.0	[-78.8, -77.2]%
7	-78.7	[-79.7, -77.6]%	-79.3	[-80.2, -78.5]%
8	-80.3	[-81.4, -79.2]%	-81.0	[-81.9, -80.1]%
9	-81.3	[-82.6, -80.0]%	-82.1	[-83.1, -81.1]%
10	-83.0	[-84.4, -81.4]%	-83.8	[-84.8, -82.7]%

Table A3: Synthetic difference-in-differences by retraction-year cohort: stacks with three complete pre- and post-retraction years, donors are the cohort's matched comparison articles, intervals from a whole-article bootstrap. The last column is the equal-weight difference-in-differences on the same matrix.

Retraction year	Retracted / donors	Synthetic DiD	95% interval	DiD
2005	10 / 40	-5.91	[-12.30, 0.48]	-6.12
2006	37 / 168	-6.55	[-9.94, -3.16]	-6.77
2007	33 / 152	-2.37	[-5.97, 1.22]	-1.96
2008	53 / 226	-3.05	[-5.14, -0.95]	-2.96
2009	97 / 392	-2.70	[-4.31, -1.08]	-2.68
2010	144 / 426	-1.57	[-2.30, -0.85]	-1.74
2011	211 / 887	-2.85	[-3.80, -1.90]	-2.88
2012	225 / 866	-2.41	[-3.38, -1.44]	-2.37
2013	257 / 1,079	-2.05	[-2.54, -1.57]	-2.26
2014	241 / 1,030	-2.58	[-3.40, -1.76]	-3.07
2015	343 / 1,426	-2.24	[-2.89, -1.59]	-2.41
2016	384 / 1,547	-1.87	[-2.35, -1.39]	-2.02
2017	392 / 1,459	-2.11	[-2.50, -1.72]	-2.08
2018	456 / 1,694	-2.72	[-3.68, -1.77]	-2.80
2019	556 / 1,930	-1.85	[-2.24, -1.45]	-1.85
2020	714 / 2,626	-2.16	[-2.51, -1.82]	-2.19
2021	1,082 / 2,924	-2.75	[-3.11, -2.38]	-2.57
2022	1,485 / 4,005	-2.58	[-2.90, -2.25]	-2.64

A Sample flow

B Event-year estimates

C Synthetic difference-in-differences by cohort

D Retraction rates and timing

E The 2016 Web of Science sample

The acknowledgment coding and the measurement comparison use a set of retracted articles and citations to them assembled from the Web of Science (WoS) in August 2016 (Analytics, 2016). We used WoS because it is one of the most comprehensive curated databases of academic research from all disciplines, and, unlike freely available databases like Google Scholar, WoS offers a way to filter results based on correction flags as well as the ability to download standardized article records containing extensive metadata.

To build the database of retracted articles, we started by creating a list of retraction notices. We searched WoS for titles containing the phrase "retraction of." The search yielded more than 14,000 records. Using the "corrections" filter in WoS—it is a WoS flag for retraction and correction notices—we filtered the list to 4,085 retraction notices. Next, we wrote software to automatically

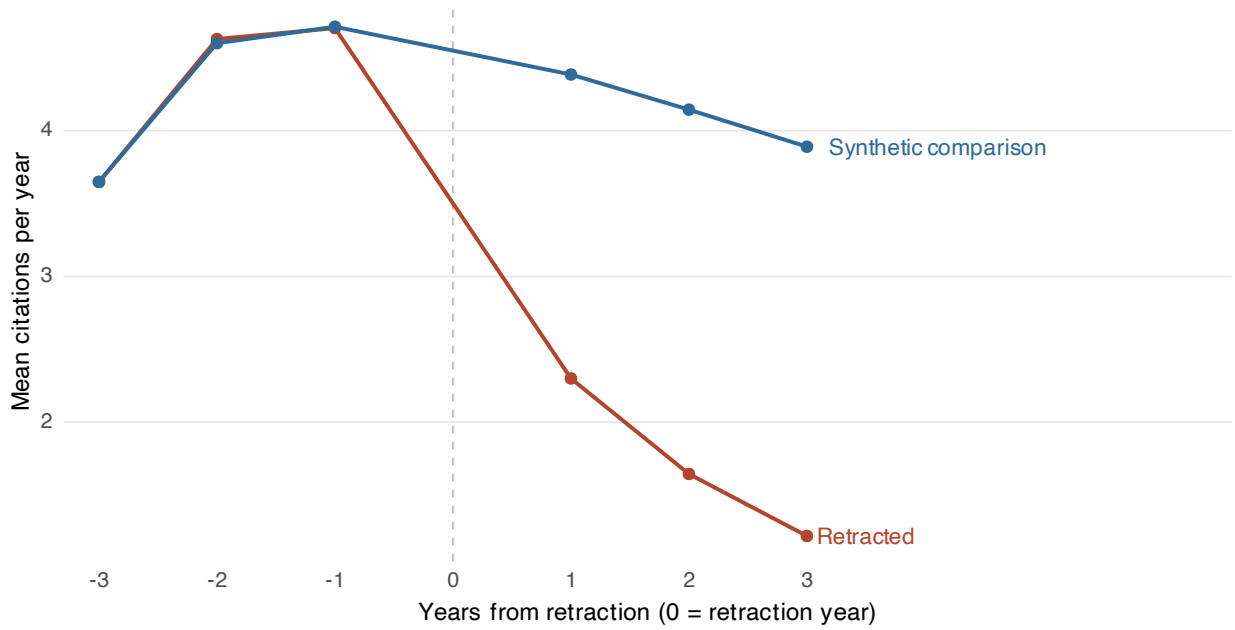


Figure A1: Mean citations per year to retracted articles and the synthetic comparison path, pooled over cohorts with weights equal to the number of retracted articles.

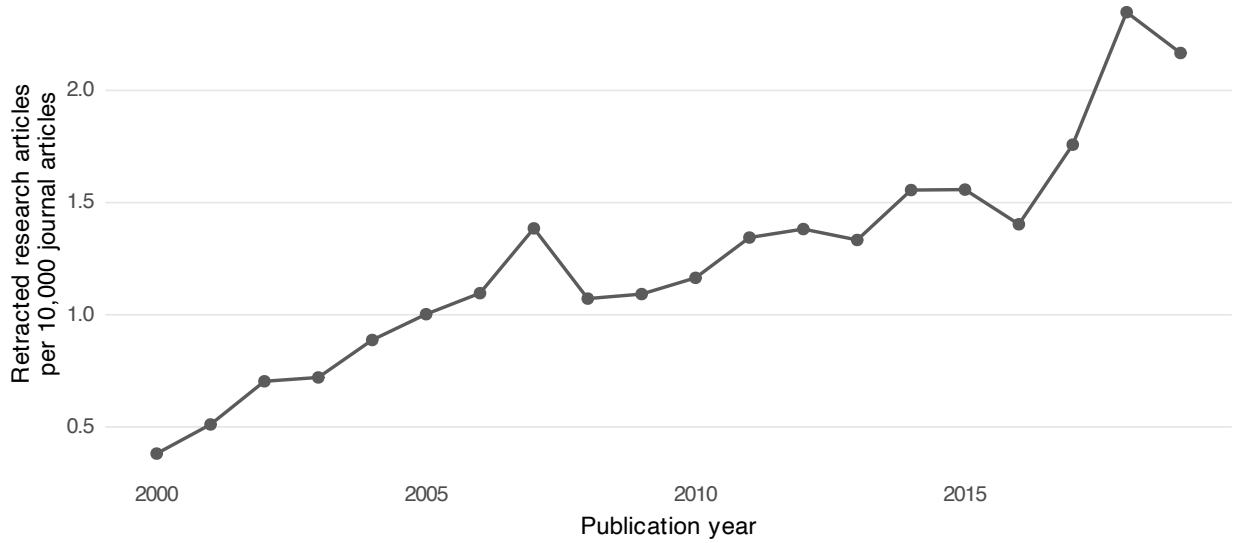


Figure A2: Frame articles per 10,000 OpenAlex journal articles, by publication year. Later cohorts have had less time to be retracted within the 2005–2022 window.

search the WoS database for retracted articles using the information in the retraction notice records. Retraction notice records did not contain consistent titles to allow a simple search, but 99% of the retraction notices contained the year the original article was published, and 96% listed the authors of the original work. We used these two pieces of information along with the name of the publication

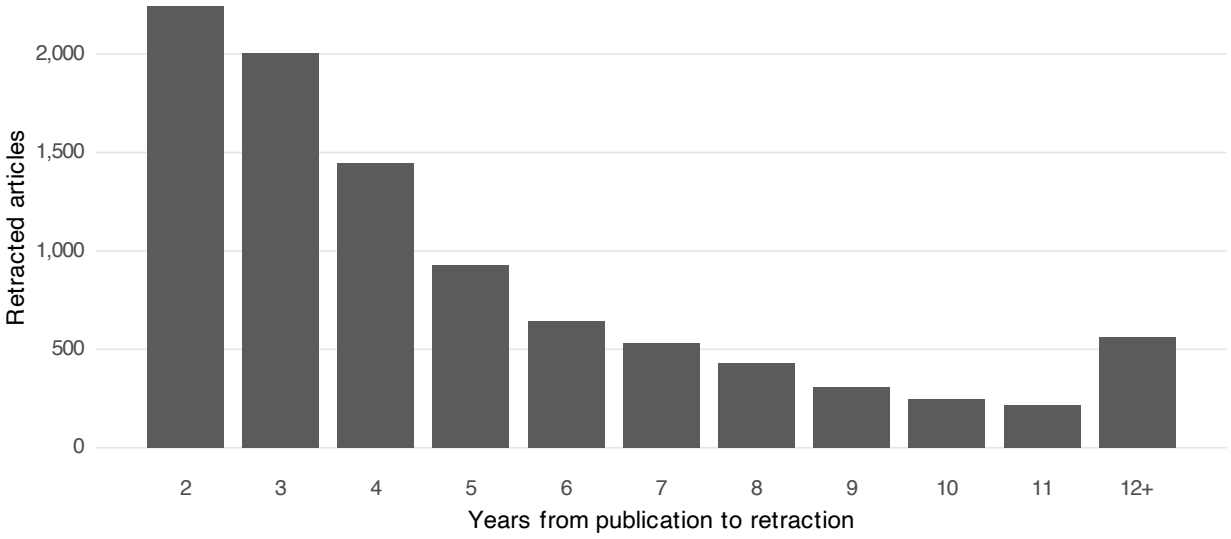


Figure A3: Years from publication to retraction for the frame articles.

to search the WoS for the original articles. The search resulted in a list of 3,776 articles. We could not locate the remaining 309 retracted articles.

Due to the variability in the information contained in the retraction notice records, the automated search process returned the wrong article in some cases. Our aim was to have zero false positives, even at the risk of some false negatives. With that aim, we created rules to flag potential false positives. First, if the list of authors of the retracted article did not match the list of authors for the relevant retraction notice record, we flagged the record as a potential false positive. Second, if the title of the retracted article did not contain the words "retracted" or "retraction," we flagged it as a potential false positive.[3] Third, we parsed the title of the retraction notices to extract the title of the original retracted article, and we flagged articles where the titles did not match as potential false positives. We then reviewed the potential false positives, filtering out all records where we could not verify the match. This resulted in a set of 3,084 articles. Finally, we checked for duplicates. We found 55. This left us with 3,029 articles. To get a list of citations to these articles, we used the WoS functionality that allows users to access the list of citations to articles and wrote software to download the citation records for each retracted article automatically. In total, we found 73,564 citations, of which 31.2% were published after the year of the retraction notice.

Table A4 compares, for the same articles, the citations per year recorded in this export with those recorded in OpenAlex.

[3]Our data suggests that it has become common practice for titles of original articles to be revised to indicate that the article has been retracted. However, adherence may vary across disciplines. As a result, the 2016 sample may be biased in favor of disciplines where the adherence is greater.

Table A4: Citations per year to the same articles in the 2016 Web of Science export and in OpenAlex, from each article's publication year through 2015. Ratio is OpenAlex over Web of Science; omitted where Web of Science recorded no citations.

Year	Article-years	Web of Science	OpenAlex	Ratio
Before 2000	236	1,083	933	0.86
2000	120	565	585	1.04
2001	180	1,032	1,051	1.02
2002	244	1,686	1,705	1.01
2003	317	1,835	1,907	1.04
2004	426	2,286	2,446	1.07
2005	555	2,808	3,040	1.08
2006	728	3,326	3,605	1.08
2007	903	3,894	4,259	1.09
2008	1,113	4,164	4,512	1.08
2009	1,363	4,863	5,252	1.08
2010	1,641	5,641	6,073	1.08
2011	1,854	6,180	6,921	1.12
2012	2,116	6,260	7,225	1.15
2013	2,339	6,123	7,011	1.15
2014	2,555	6,092	6,934	1.14
2015	2,661	5,127	6,002	1.17

F Coding citations as acknowledging problems or not

To code the citations, we downloaded citing articles and their associated retracted article. A research assistant then edited the citing article pdf to highlight where the retracted article was discussed in the citing article. The judgment of whether the article noted any concerns was made based on a review of the original retracted article pdf and the highlighted text.

If an article did not note any concerns with the cited article, it was coded as *not acknowledging problems*. Simply disagreeing with the conclusions of an article without noting any concern still meant that the article was being cited in a way that suggests that its findings are trustworthy and was also coded as *not acknowledging problems*. We code articles that note any concern with the cited article, even those unrelated to the cause of retraction, as *acknowledging problems*. We evaluated the reliability of the coding by having an independent rater code 50 randomly selected citing articles. The two sets of independent codes were found to agree in all 50 instances.